2. Second-Order Runge–Kutta Methods

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Euler's method uses a single slope, taken at the start of the step. **Runge–Kutta methods** evaluate f at a few trial points inside the step (called **stages**) and combine the slopes into a better estimate of the average slope. This section covers the methods with two stages: Heun, midpoint and Ralston. All three have order 2.

Test problem: $y' = x + y$, $y(0) = 1$, exact solution $y = 2e^x - x - 1$.

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (7, 4.5)
plt.rcParams["axes.grid"] = True
np.set_printoptions(precision=6, suppress=True)

f = lambda x, y: x + y
exact = lambda x: 2 * np.exp(x) - x - 1

def euler(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        y[i + 1] = y[i] + h * f(x[i], y[i])
        x[i + 1] = x[i] + h
    return x, y
```

2.1 Heun's method (improved Euler)

Derivation from the trapezoidal rule. Estimate the integral in $y(x_{n+1}) = y(x_n) + \int_{x_n}^{x_{n+1}} f\,dx$ with the trapezoidal rule, which averages the slopes at both ends of the step:

$$y_{n+1} = y_n + \frac{h}{2}\Big[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\Big].$$

The problem is that y_{n+1} appears on both sides (the formula is **implicit**). Heun's fix is to first *predict* y_{n+1} with an Euler step and then *correct* it with the trapezoidal average. That makes it a **predictor–corrector** method:

$$\begin{aligned} k_1 &= f(x_n, y_n) && \text{slope at the start} \\ y^*_{n+1} &= y_n + hk_1 && \text{predictor (an Euler step)} \\ k_2 &= f(x_{n+1}, y^*_{n+1}) && \text{slope at the predicted end point} \\ y_{n+1} &= y_n + \tfrac{h}{2}(k_1 + k_2) && \text{corrector: use the average slope} \end{aligned}$$

Why it is better. If the curve is concave up, the start slope k_1 is too small and the end slope k_2 is too large, so their average largely cancels the error. The local error becomes $O(h^3)$ and the global error $O(h^2)$, so Heun is **order 2**. Halving h divides the error by about 4. Each step costs **2** evaluations of f.

```
def heun(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
        k2 = f(x[i] + h, y[i] + h * k1)
        y[i + 1] = y[i] + h / 2 * (k1 + k2)
        x[i + 1] = x[i] + h
    return x, y
```

Worked example: Heun by hand, $h = 0.1$

n	x_n	y_n	$k_1 = x_n + y_n$	$y^* = y_n + 0.1k_1$	$k_2 = x_{n+1} + y^*$	$y_{n+1} = y_n + 0.05(k_1 + k_2)$	error
0	0.0	1.000000	1.000000	1.100000	1.200000	1.110000	0.00034
1	0.1	1.110000	1.210000	1.231000	1.431000	1.242050	0.00076
2	0.2	1.242050	1.442050	1.386255	1.686255	1.398465	0.00125

After three steps the error is 0.00125, about **30 times smaller** than Euler's 0.0377 (Section 1.2), for twice the work.

2.2 The midpoint method

Derivation from the midpoint rule. Approximate the integral by h times the slope at the **middle** of the step, $x_n + h/2$. The value of y there is unknown, so estimate it with half an Euler step:

$$\begin{aligned} k_1 &= f(x_n, y_n) \\ k_2 &= f\big(x_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}k_1\big) && \text{slope at the estimated midpoint} \\ y_{n+1} &= y_n + h\,k_2 \end{aligned}$$

Like Heun, it uses two evaluations of f per step and has order 2.

```
def midpoint(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
```

```
        k2 = f(x[i] + h / 2, y[i] + h / 2 * k1)
        y[i + 1] = y[i] + h * k2
        x[i + 1] = x[i] + h
    return x, y
```

2.3 A whole family: deriving the conditions for order 2

Heun and midpoint are two members of a one-parameter family. Consider any method of the form

$$k_1 = f(x_n, y_n), \qquad k_2 = f(x_n + c\,h,\ y_n + c\,h\,k_1),$$
$$y_{n+1} = y_n + h\,(b_1 k_1 + b_2 k_2),$$

and ask: which b_1, b_2, c make the method agree with the Taylor series of the exact solution for as long as possible?

Taylor series of the exact solution. Differentiating $y' = f(x, y)$ with the chain rule gives $y'' = f_x + f_y\,y' = f_x + f f_y$, where subscripts denote partial derivatives evaluated at (x_n, y_n). So

$$y(x_n + h) = y + hf + \frac{h^2}{2}(f_x + f f_y) + O(h^3).$$

Taylor series of the method. Expand k_2 in two variables:

$$k_2 = f(x + ch,\ y + chf) = f + ch\,f_x + chf\,f_y + O(h^2),$$

so

$$y_{n+1} = y + h\,(b_1 + b_2)\,f + h^2\,b_2 c\,(f_x + f f_y) + O(h^3).$$

Matching the h and h^2 terms for every f requires

$$\boxed{b_1 + b_2 = 1, \qquad b_2\,c = \tfrac{1}{2}}$$

Two equations for three unknowns: one free parameter c, with $b_2 = \frac{1}{2c}$ and $b_1 = 1 - \frac{1}{2c}$.

c	b_1	b_2	Method
1	$\frac{1}{2}$	$\frac{1}{2}$	Heun
$\frac{1}{2}$	0	1	midpoint
$\frac{2}{3}$	$\frac{1}{4}$	$\frac{3}{4}$	Ralston

Why two stages cannot reach order 3. The h^3 term of the exact solution contains $y''' = f_{xx} + 2f f_{xy} + f^2 f_{yy} + f_y(f_x + f f_y)$. The method's h^3 term comes only from the second-order part of the expansion of k_2, which contains f_{xx}, f_{xy}, f_{yy} but **never** the product $f_y(f_x + f f_y)$. No choice of b_1, b_2, c can match it, so order 3 needs at least three stages.

Ralston's choice $c = \frac{2}{3}$ minimises a bound on the h^3 error term over all f, so it is often slightly more accurate than Heun or midpoint.

The code implements the whole family with c as a parameter.

```
def rk2(f, x0, y0, h, n, c):
    b2 = 1 / (2 * c)
    b1 = 1 - b2
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
        k2 = f(x[i] + c * h, y[i] + c * h * k1)
        y[i + 1] = y[i] + h * (b1 * k1 + b2 * k2)
        x[i + 1] = x[i] + h
    return x, y

ralston = lambda f, x0, y0, h, n: rk2(f, x0, y0, h, n, 2 / 3)

# rk2 with c = 1 and c = 1/2 must reproduce Heun and midpoint exactly
print(np.max(np.abs(rk2(f, 0, 1, 0.1, 10, 1.0)[1] - heun(f, 0, 1, 0.1, 10)[1])))
print(np.max(np.abs(rk2(f, 0, 1, 0.1, 10, 0.5)[1] - midpoint(f, 0, 1, 0.1, 10)[1])))
```

```
0.0
0.0
```

2.4 Comparing the family

A caution about the test problem. For $f = x + y$ every member of the family gives the *same* numbers. With $k_1 = x_n + y_n$,

$$k_2 = (x_n + ch) + (y_n + chk_1) = k_1 + ch(1 + k_1), \qquad y_{n+1} = y_n + hk_1 + \underbrace{b_2 c}_{=1/2}\, h^2(1 + k_1),$$

which does not depend on c. Because f is linear, the second derivatives f_{xx}, f_{xy}, f_{yy} vanish and the only place the methods could differ disappears. To compare them we need an f that is **nonlinear** in x or y:

$$y' = y - x^2 + 1, \qquad y(0) = 0.5, \qquad \text{exact solution } y = (x+1)^2 - \tfrac{1}{2}e^x.$$

```
p = lambda x, y: y - x**2 + 1
p_exact = lambda x: (x + 1)**2 - 0.5 * np.exp(x)

methods = {"Euler": euler, "Heun": heun, "Midpoint": midpoint, "Ralston": ralston}
hs = [0.2, 0.1, 0.05, 0.025]
print(f"error at x = 2 for y' = y - x² + 1\n{'h':>7}" + "".join(f"{m:>12}{'ratio':>7}" for m in methods))
errs = {m: [] for m in methods}
for i, h in enumerate(hs):
    row = f"{h:7.3f}"
    for m, method in methods.items():
        _, y = method(p, 0, 0.5, h, round(2 / h))
        errs[m].append(abs(p_exact(2) - y[-1]))
        ratio = f"{errs[m][-2] / errs[m][-1]:7.2f}" if i else " " * 7
        row += f"{errs[m][-1]:12.3e}{ratio}"
    print(row)
```

```
error at x = 2 for y' = y - x² + 1
      h       Euler  ratio        Heun  ratio    Midpoint  ratio     Ralston  ratio
  0.200   4.397e-01          7.242e-02          1.510e-02          3.421e-02
  0.100   2.420e-01   1.82   1.890e-02   3.83   3.747e-03   4.03   8.800e-03   3.89
```

```
  0.050    1.275e-01    1.90    4.820e-03    3.92    9.277e-04    4.04    2.225e-03    3.95
  0.025    6.550e-02    1.95    1.216e-03    3.96    2.304e-04    4.03    5.590e-04    3.98
```

The ratios approach 2 for Euler and 4 for the three second-order methods. On this problem the midpoint method has the smallest error and Heun the largest, with Ralston in between. Ralston's choice minimises a bound over *all* f, which does not make it the best for every particular f. All three are far ahead of Euler.

2.5 Seeing the slopes

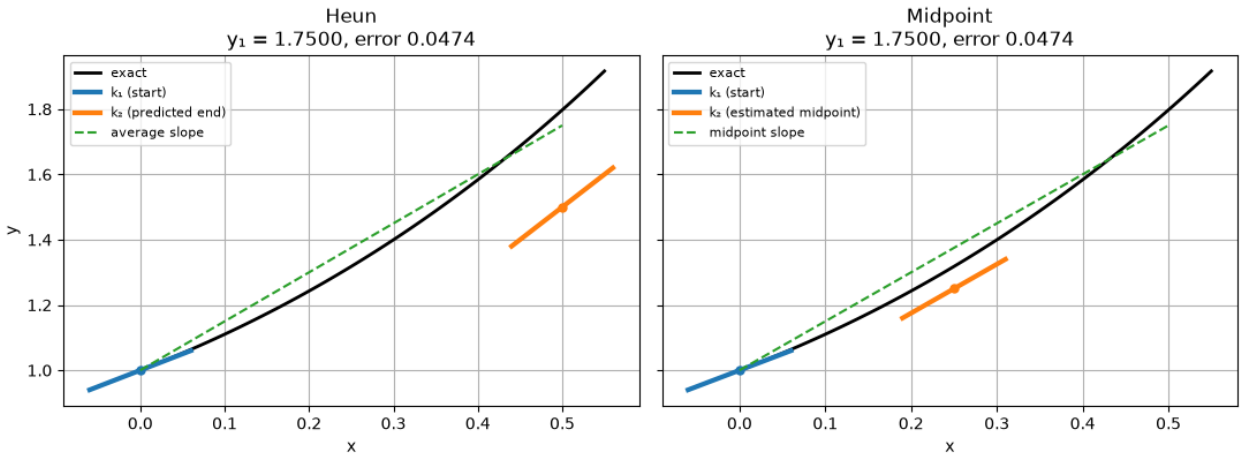
One large step ($h = 0.5$) on the test problem shows where each method samples the slope. Heun samples at the **end** of the step, midpoint in the **middle**. Short thick lines mark each sampled slope; the dashed line is the step actually taken.

```python
def slope_mark(ax, x, y, s, color, label=None):
    d = 0.06
    ax.plot([x - d, x + d], [y - d * s, y + d * s], color=color, lw=3, label=label)
    ax.plot(x, y, "o", color=color, ms=5)

h, x0, y0 = 0.5, 0.0, 1.0
xf = np.linspace(0, 0.55, 100)
fig, axes = plt.subplots(1, 2, figsize=(11, 4.2), sharey=True)
for ax in axes:
    ax.plot(xf, exact(xf), "k", lw=2, label="exact")
    ax.set_xlabel("x")
axes[0].set_ylabel("y")

k1 = f(x0, y0)
k2 = f(x0 + h, y0 + h * k1)
y_h = y0 + h / 2 * (k1 + k2)
slope_mark(axes[0], x0, y0, k1, "C0", "k  (start)")
slope_mark(axes[0], x0 + h, y0 + h * k1, k2, "C1", "k  (predicted end)")
axes[0].plot([x0, h], [y0, y_h], "C2--", label="average slope")
axes[0].set_title(f"Heun\ny  = {y_h:.4f}, error {exact(h) - y_h:.4f}")

m2 = f(x0 + h / 2, y0 + h / 2 * k1)
y_m = y0 + h * m2
slope_mark(axes[1], x0, y0, k1, "C0", "k  (start)")
slope_mark(axes[1], x0 + h / 2, y0 + h / 2 * k1, m2, "C1", "k  (estimated midpoint)")
axes[1].plot([x0, h], [y0, y_m], "C2--", label="midpoint slope")
axes[1].set_title(f"Midpoint\ny  = {y_m:.4f}, error {exact(h) - y_m:.4f}")
for ax in axes:
    ax.legend(loc="upper left", fontsize=8)
plt.tight_layout()
```



The two methods sample different slopes but land on the same value, as Section 2.4 predicted for this linear test problem.

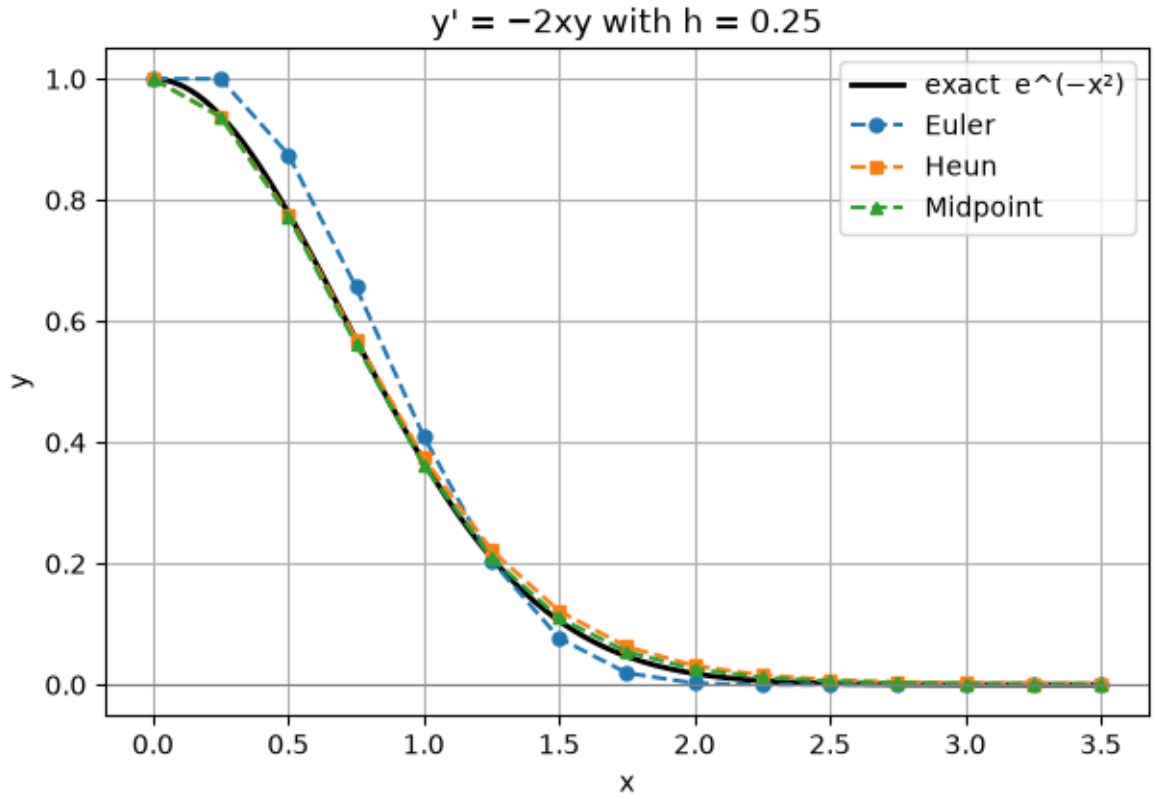
2.6 Example: a problem where Euler misbehaves

$$y' = -2xy, \qquad y(0) = 1, \qquad \text{exact solution } y = e^{-x^2}, \qquad h = 0.25.$$

Euler's update is $y_{n+1} = (1 - 2x_n h)\, y_n$. At $x_n = 2$ the factor is exactly 0, and for $x_n > 2$ it is **negative**, so Euler's solution hits zero and then changes sign at every step. The true solution is positive and decays smoothly. (Section 5 explains this as an instability: the step is too large for how quickly the solution decays.)

```
g = lambda x, y: -2 * x * y
g_exact = lambda x: np.exp(-x**2)
h, n = 0.25, 14
xf = np.linspace(0, 3.5, 300)
plt.plot(xf, g_exact(xf), "k", lw=2, label="exact  e^(-x²)")
for name, method, style in [("Euler", euler, "o--"), ("Heun", heun, "s--"), ("Midpoint", midpoint, "^--"
    x, y = method(g, 0, 1, h, n)
    plt.plot(x, y, style, ms=5, label=name)
    print(f"{name:8s} max |error| = {np.max(np.abs(y - g_exact(x))):.4f}")
plt.axhline(0, color="gray", lw=0.8)
plt.xlabel("x"); plt.ylabel("y"); plt.title("y' = -2xy with h = 0.25"); plt.legend();
```

```
Euler    max |error| = 0.0962
Heun     max |error| = 0.0161
Midpoint max |error| = 0.0084
```



2.7 Summary

Method	Stages (f evaluations per step)	Order	Idea
Euler	1	1	slope at the start
Heun	2	2	average of start slope and predicted end slope
Midpoint	2	2	slope at an estimated midpoint
Ralston	2	2	the member of the family with the smallest error bound

- Every two-stage method with $b_1 + b_2 = 1$ and $b_2 c = \frac{1}{2}$ has order 2. Two stages can never give order 3.
- On a linear test problem all of them agree. Use a nonlinear problem to tell them apart.

Exercises

1. For $y' = y - x^2 + 1$, $y(0) = 0.5$, take one step with $h = 0.2$ using Heun, midpoint and Ralston by hand. Compare with $y(0.2) = 1.44 - 0.5e^{0.2}$.
2. Show that for $y' = x + y$, $y(0) = 1$, the first step of any member of the family gives $y_1 = 1 + h + h^2$. How does this compare with the Taylor series $y(h) = 1 + h + h^2 + h^3/3 + \cdots$?
3. Apply a two-stage method to $y' = 1$. Show that it gives the exact answer only if $b_1 + b_2 = 1$. (This is why that condition is called **consistency**.)
4. Write down the member of the family with $c = \frac{3}{4}$ and measure its order numerically.

Answers

1. Heun: $k_1 = 1.5$, $y^* = 0.8$, $k_2 = 1.76$, $y_1 = 0.826$. Midpoint: $k_2 = f(0.1, 0.65) = 1.64$, $y_1 = 0.828$. Ralston: $k_2 = f(0.1\overline{3}, 0.7) = 1.682\overline{2}$, $y_1 = 0.827\overline{3}$. Exact: 0.829299.
2. $k_1 = 1$, $k_2 = 1 + ch(1 + 1) = 1 + 2ch$, so $y_1 = 1 + h(b_1 + b_2) + 2b_2 c\, h^2 = 1 + h + h^2$. It matches the Taylor series up to h^2 and misses $h^3/3$: that is the local error.
3. $k_1 = k_2 = 1$, so $y_{n+1} = y_n + (b_1 + b_2)h$, while the exact solution increases by h.
4. $b_2 = \frac{2}{3}$, $b_1 = \frac{1}{3}$. See the cell below: the error ratio approaches 4.

```python
for name, method in [("Heun", heun), ("Midpoint", midpoint), ("Ralston", ralston)]:
    _, y = method(p, 0, 0.5, 0.2, 1)
    print(f"1.  {name:8s} y(0.2)   {y[1]:.6f}   (exact {p_exact(0.2):.6f})")

prev = None
for h in [0.2, 0.1, 0.05, 0.025]:
    _, y = rk2(p, 0, 0.5, h, round(2 / h), c=3 / 4)
    err = abs(p_exact(2) - y[-1])
    print(f"4.  c = 3/4, h = {h:5}: error {err:.3e}" + (f",  ratio {prev / err:.2f}" if prev else ""))
    prev = err
```

```
1.  Heun     y(0.2)   0.826000   (exact 0.829299)
1.  Midpoint y(0.2)   0.828000   (exact 0.829299)
1.  Ralston  y(0.2)   0.827333   (exact 0.829299)
4.  c = 3/4, h =   0.2: error 4.376e-02
4.  c = 3/4, h =   0.1: error 1.133e-02,  ratio 3.86
4.  c = 3/4, h =  0.05: error 2.874e-03,  ratio 3.94
4.  c = 3/4, h = 0.025: error 7.233e-04,  ratio 3.97
```